

Python Operators :

$x + y$
 $x - y$
 $x * y$

→ operators

Types of Operators :

1. Arithmetic Operators
2. Assignment Operators
3. Relational Operators
4. Logical Operators
5. Bitwise Operators ?
6. Identity Operators ?

1. Arithmetic Operators

$+$ → Addition

$-$ → Subtraction

$*$ → Multiplication

$/$ → division

$//$ → floor division

$\%$ → modulus operator

$**$ → Exponentiation → left most

round off value → nearest round off value

$10 // 3 = 3$
$10 / 3 = 3.33$

modulus ($\%$) :-

→ It gives remainder as result

$10 \div 3 = 3$

→ It gives remainder as result

Ex- $10 \% 3 = 1$

$10 \% 5 = 0$

Note:- $a \% b$

If $a < b$
↓
ans = a
↙
 $2 \% 5 = 2$

$$\begin{array}{r} 3 \overline{) 10} \\ - 9 \\ \hline 1 \end{array}$$

$$\begin{array}{r} 5 \overline{) 10} \\ - 10 \\ \hline 0 \end{array}$$

Exponentiation:-

$2^2 = 4 \Rightarrow 2 * 2$

$2^3 = 8 \Rightarrow 2 * 2 * 2$

$3^3 = 27 \Rightarrow 3 * 3 * 3$

$10^3 = 1000 \Rightarrow 10 * 10 * 10$

2. Assignment Operators → It is combination of arithmetic & equals operators

+=

-=

*=

/=

//=

%=

**=

$x = 10$

$x += 1$

$\text{print}(x)$

$0/P \Rightarrow 11$

$x += 1 \rightarrow x = x + 1$

$x = 10 + 1 \Rightarrow 11$

$x = 10$

$x -= 5$

$x -= 5 \rightarrow x = x - 5$

$x = 10$

$x -= 5$

`print(x)`

$0 \mid P \Rightarrow 5$

$x -= 5 \rightarrow x = x - 5$

$x = 10 - 5 = 5$

3. Relational Operators $\rightarrow$ output of this either TRUE OR FALSE

$== \rightarrow$ equals to

$>$

$<$

$\geq$

$\leq$

$!=$

$x = 10 \quad y = 10 \quad x == y \rightarrow \text{True}$

$x = 2 \quad y = 10 \quad x == y \rightarrow \text{False}$

$a = 10 \quad b = 11$

$a > b \rightarrow 10 > 11 \times \text{False}$

$x = 10$

$y = 2$

$x \geq 2 \rightarrow x > 2 \text{ OR } x == 2$
 $\downarrow$
True

$!= \rightarrow$ not equals $\sim$ give True when values are not equal otherwise False.

$x = 10$

$y = 5$

$x != y \rightarrow \text{True}$

$x = 7$

$x \geq 2 \rightarrow \text{True}$

$x = 2$

$x \geq 2$

$\downarrow$
True

$x = 10$

$y = 10$

$x != y \rightarrow \text{False}$

4. Logical Operators

AND

OR

both are true $\rightarrow$ then TRUE

$x > 2$ and $x > 10$
 $x > 2$ OR $x > 10$

either one is true $\rightarrow$ then true

Home Work :

1. Armstrong Number : [Sandbox: A Kid-Friendly Platform for Coding & Math | Codeyoung](#)
2. Even or Odd : [Sandbox: A Kid-Friendly Platform for Coding & Math | Codeyoung](#)